PV+BESS Auction Trading: Executed Implementation Status

Repository implementation evidence

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1 Snapshot and scope

This report describes delivered code and observed executions, with machine-readable evidence in `implementation_status.json`. The repository is `/home/rghetti/bess_pv_trading`; revision `839793674af5` identifies the base revision at report generation; the accompanying JSON records the working-tree state used for this snapshot. The original audit modules and raw stores remain intact. A compatibility bridge exposes their existing imports alongside the new `src/` application.

The solved system is one AC PV farm with an AC-coupled lithium-ion battery and one POI, using guaranteed fixed-quantity DA/IDA1/IDA2/IDA3 energy adjustments. Every submitted quantity executes at its auction clearing price; that price becomes known at publication. Each model contains all represented scenarios with node-owned decisions. No portfolio POI, continuous trading, ancillary services or price impact is modeled. Each run is an independent single-delivery-day study. Current farm weather has one deterministic member; multiple prices constitute *stochastic-price / deterministic-PV*.

Python version: `3.14.8 | packaged by conda-forge | (main, Oct 12 2026, 17:35:27) [GCC 15.3.0]`. Imported library versions used: numpy 2.5.4, pandas 3.0.6, scipy 1.18.1, gurobipy 13.0.3, pvlib 0.16.1, xarray 2026.7.0, zarr 2.18.7, pyarrow 25.0.0, PyYAML 6.0.3, pytest 9.1.1, matplotlib 3.11.2. Scikit-learn was unavailable; Ridge uses NumPy linear algebra. No packages, environments or licenses were changed. A tiny Gurobi model failed with sandbox HostID masking and succeeded outside the sandbox; licensed solver tests and experiments used the latter execution context.

2 Component status and evidence

Module paths and exact test identities are recorded in the JSON evidence. The status below distinguishes tested software from unavailable historical inputs.

Component	Status	Executed evidence or blocker
Deterministic and stochastic MILP `src/bess_pv_trading/optimization/model.py`	implemented-and-tested	Licensed analytical/unit tests; ordinary and DST run manifests
Nonanticipativity and explicit information tree `src/bess_pv_trading/domain.py; scenarios.py`	implemented-and-tested	Bid/control owner tests, unresolved descendants, duplicate-observation grouping
Masking, interpolation and evidence as-of views `src/bess_pv_trading/data/policy.py`	implemented-and-tested	Policy fixtures and selected-farm actual slice manifest

Component	Status	Executed evidence or blocker
Read-only farm adapter `src/bess_pv_trading/data/adapters.py`	implemented-and-tested	Actual Zarr metadata and labeled Parquet joins
pvlib AC forecast and calibration interface `src/bess_pv_trading/forecasting/pv.py`	implemented-and-tested	Fixed/tracker fixtures; actual forecast has flagged invalid intervals
Actual full-day PV eligibility `src/bess_pv_trading/forecasting/pv.py`	partial	4 invalid intervals in the executed real-data example; no imputation
Manual GME CSV/mapped XML import and cache `src/bess_pv_trading/data/market_prices.py`	implemented-and-tested	Synthetic canonical CSV import/cache-hit and gaps; mapped XML tests
GME API request and zonal-price parser `src/bess_pv_trading/data/gme.py; data/api.py`	implemented-and-tested	Documented MGP/MI-A1/2/3 contracts; offline authentication, JSON ZIP, native-product and cache tests; live coverage pending
GME authenticated historical acquisition `src/bess_pv_trading/data/gme.py`	blocked	User account access and authenticated download evidence remain pending; no actual GME IDA prices are claimed
Prepared forecast-to-optimization adapter `src/bess_pv_trading/optimization/prepared.py`	implemented-and-tested	Initial pre-DA point planning; complete/causal CSV and missing-input rejection tests
Artifact-to-stochastic optimization input boundary `src/bess_pv_trading/optimization/forecast_inputs.py`	implemented-and-tested	Matching JUnit fixture execution; joint price/PV members, known results, guaranteed commitments, residual grids and IDA3 eligibility; no actual-market economic backtest
Terna API fundamentals and OAuth `src/bess_pv_trading/data/terna.py;data/api.py`	implemented-and-tested	Verified provider schemas; offline OAuth/cache/parser tests; live downloads left to the user
ENTSO-E API prices and forecasts `src/bess_pv_trading/data/entsoe.py;data/api.py`	implemented-and-tested	A44 DA/native product and forecast parser tests; IDA/imbalance not assumed
Monthly downloads and offline cache catalog `src/bess_pv_trading/data/api.py; data/api_catalog.py`	implemented-and-tested	All requested datasets per local month; unchanged daily cache keys; readiness retains no-data gaps
Consolidated DuckDB API storage `src/bess_pv_trading/data/duckdb_store.py`	implemented-and-tested	Transactional raw/normalized snapshots, SQL records and retained provenance; new mean-model run ingested DuckDB snapshots
Ridge level/spread fits and OOS errors `src/bess_pv_trading/forecasting/prices.py`	implemented-and-tested	Four synthetic CLI-trained models; causal lag and rolling-origin tests
Retrospective DA price experiments `src/bess_pv_trading/forecasting/price_experiment.py`	implemented-and-tested	Configured retrieved-revision experiment manifest; native durations and chronological hold-out
Gaussian/bootstrap residuals and fitted tree interface `src/bess_pv_trading/forecasting/prices.py;scenarios.py`	implemented-and-tested	Legacy causal fixture interface; retrieved-revision calibration is reported separately

Component	Status	Executed evidence or blocker
Regional forecast feature comparison `src/bess_pv_trading/forecasting/fundamental_experiment.py`	implemented-and-tested	January retrieved-revision load/solar/wind comparison; complete intervals and shared holdout
Rolling linear price selection and forecast artifacts `src/bess_pv_trading/forecasting/linear_price_study.py; forecasting/price_features.py`	implemented-and-tested	Actual retrieved-revision studies; validation-only specification choice, weekly refits and independent later calibration/holdout
Alternative standardized linear price estimators `src/bess_pv_trading/forecasting/linear_estimators.py`	implemented-and-tested	OLS, recency-weighted ridge, robust Huber IRLS, lasso and elastic-net; training-only scalers, saved-predictor and convergence tests
Zero-mean Gaussian paths and MILP adapter `src/bess_pv_trading/forecasting/gaussian_prices.py;forecasting/price_pipeline.py`	implemented-and-tested	Separate OOS bias/error covariance, native/DST grids and scenario-contract tests; historical deployment remains ineligible
Calibration-selected online Gaussian uncertainty `src/bess_pv_trading/forecasting/gaussian_adaptive.py; forecasting/gaussian_study.py`	implemented-and-tested	Frozen inner-calibration policy; only earlier complete mature errors update daily bias/covariance; static comparison and scored coverage
Actual-price perfect-information scenario benchmark `src/bess_pv_trading/forecasting/price_oracle.py`	implemented-and-tested	Deterministic actual clearing prices per PV member; latent PV histories, native labels and licensed analytical upper-bound tests
Historical learned price/PV scenarios `src/bess_pv_trading/forecasting/prices.py`	blocked	Verified historical vintages, IDA labels, causal multiauction calibration and settlement unavailable
Commitment ledger and synthetic replay `src/bess_pv_trading/simulation/synthetic.py`	implemented-and-tested	Guaranteed full execution; quantities fixed at submission, actual cash booked once at publication; four benchmarks and 92/96/100 product days
Historical market backtest `src/bess_pv_trading/cli.pybacktest`	blocked	Requires complete causal auction/settlement inputs, eligible PV and confirmed BESS/POI
Constant throughput aging `src/bess_pv_trading/optimization/model.py`	implemented-and-tested	AC charge-plus-discharge convention and carried cumulative cost
Xu cycle-depth segmented aging `src/bess_pv_trading/optimization/degradation.py`	implemented-and-tested	Eq.5/7 segment costs, appendix priority/state carry; analytical/rainflow/solver/replay tests
Independent numerical validation `src/bess_pv_trading/optimization/validation.py`	implemented-and-tested	Recomputed accounting/bounds; intentionally altered solutions rejected

3 Dataset contract and derived observations

The verified files are `data/farms/pv_farms_timeseries_20250101_20260831.zarr` and the matching `pv_farms_timeseries_20250101_20260831.parquet`. Metadata inspection observed four farms, 58,368 historical timestamps per farm, 2,440 forecast cycles, and 49 hourly leads. Observations cover

1 January 2025 through 31 August 2026 in UTC; forecast boundary coverage differs. Leads are timedeltas from zero to 48 hours. Installed DC capacities are 953.28, 1524.60, 4160 and 6638 kW for IDs 1, 345, 901 and 910 respectively. Each is a separate plant. The measured `power` variable is already AC kW, regardless of obsolete DC column labels retained inside the unchanged audit. These meanings follow the user's corrected contract and local metadata; public documentation cannot verify private source provenance.

Power has dimensions `(time,farm_id)`; satellite GHI and diffuse fraction use `(farm_id,time)`. Forecasts use `(farm_id,cycle_time,lead_time)`. Named transposes and farm-label joins prevent positional assumptions. Overlapping vintages are not ensemble members. Hourly shortwave radiation is forecast GHI, while the separate quarter-hour GHI is historical satellite data. The direct-radiation-as-BHI meaning is a working assumption. `fraction_diffuse` remains disabled as a model input.

The versioned policy first detects native nonzero anchor-matched power runs of at least four samples, using assumed tolerances 0.01 kW absolute and 0.0001 relative, and zeroes the entire qualifying run. Zero, missing value or timestamp gap breaks a run. It then zeroes positive production from 21:30 inclusive through 05:00 exclusive in Europe/Rome, floors only allowlisted negative forecast magnitudes, and linearly interpolates internally bracketed missing historical power/GHI. The default has no maximum interior gap. It neither extrapolates nor fills forecast gaps, and it reapplies the production night window. Temperatures, signed wind and prices retain valid negative values. Original values, reason flags, run/gap lengths, evidence times and energy changes are exported.

The executed farm-1 UTC slice on 1 October 2025 contains 192 long-format rows. Adjusted rows: 2; remaining missing: 0. Raw/treated finite AC energies were 2.255149982918054 and 2.255149982918054 MWh, with 0.0 MWh change. This slice interpolated GHI while AC production energy was unchanged. Detailed counts and daily raw/treated energies are retained in the analysis manifest. Policy treatment is an analysis choice, not independent verification of faults or unconstrained PV availability. Real clipping plateaus may be removed. No upstream review file was configured; a missing review file is not a completed review.

4 Time semantics and causal information

Each historical label starts a 15-minute interval; its final average is available at interval end plus configured telemetry latency. Retroactive plateau evidence for the first three samples arrives only at the fourth sample's availability; later flags arrive no earlier than their sample. A two-sided interpolation also waits for its right endpoint and applicable mask evidence. As-of observation views reconstruct what was available at a cutoff; finalized treated observations are used only for labeled ex-post accounting.

Forecast availability defaults to cycle plus four hours. At the executed cutoff 2025-09-30T09:45:00+00:00, cycle 2025-09-30T00:00:00+00:00 was selected, available at 2025-09-30T04:00:00+00:00. The 06:00 cycle was not yet available. End-labeled radiation at 12:00 describes 11:00–12:00; uniform redistribution conserves its irradiation over the four contained quarters. Instantaneous weather uses within-vintage interpolation at interval midpoints. No structural forecast gap is replaced with measurements or another future cycle.

The calendar uses Europe/Rome local gates: DA D-1 12:00, IDA1 D-1 15:00, IDA2 D-1 22:00, IDA3 D 10:00, with a five-minute computation buffer. IDA3 products start at noon. IDA results use documented publication deadlines; DA 13:30 is a conservative study assumption because the rule states an earliest final time, not a guaranteed deadline. Recorded timestamps override schedules and cancellation can remove an auction. Rule revisions 13 and 14 are dated separately [4, 5]. Native quarter-hour MGP delivery begins 1 October 2025 [3]. Earlier hourly DA products remain native in retrospective price analysis; mapping them to quarter-hour dispatch is not implemented. Local days have 92, 96 or 100 products, all represented internally with aware UTC clocks.

5 Weather-to-AC implementation

The installed pvlib implementation uses solar position, total plane-of-array irradiance, `irradiance.dni`, `tracking.singleaxis`, Faiman temperature and `pvsystem.pvwatts_dc` [8, 10, 9]. Wind is the magnitude of signed 10-m components, explicitly a module-wind proxy. The limited-metadata DC model uses nameplate watts and a configured temperature coefficient, returning diagnostic MW. The example applies a separate assumed, unfitted PV-output factor 0.96. An AC calibration helper enforces chronological availability; no real calibration fit was claimed. Nameplate is not an automatic hard output cap.

Milan coordinates are acknowledged proxy geometry, not fault evidence. Fixed surfaces retain their own tilt/azimuth; single-axis farms use separately assumed axis tilt/azimuth, rotation, GCR and backtracking. Farm 901 explicitly omits rear gain under a front-side-only approximation. Section 8.2 of the user's prompt confirms west=-90 degrees; the configurable east-positive conversion and its alternative are tested. No undocumented module technology is inferred.

The BHI working interpretation computes DHI=GHI-BHI and converts to DNI with representative quarter-hour zenith, suppressing low-sun direct irradiance and flagging an explicit DNI cap. This is an interval-mean geometry approximation. Contradictory BHI exceeding GHI invalidates intervals under the example policy. The actual forecast produced 96 products with 4 invalid intervals. Its finite-only energy is not a complete-day energy estimate. Ex-post evaluation uses exactly matched measured/policy-treated AC intervals, with coverage and exclusions. On 92 eligible pairs, AC MAE and RMSE were 0.08226337562003606 and 0.12986883499437865 MW respectively. These are an unfitted one-vintage comparison, not held-out calibrated skill. The separately configured radiation sensitivity completed 96 products while retaining the contradiction flags; its detailed numerical metrics are included in JSON evidence.

Measured historical AC power is divided by 1000 to MW without another conversion. The separately configured battery converter uses 0.97 per direction, giving 0.9409 converter-only round-trip efficiency. Direct PV exports bypass it. The final positive AC forecast receives the same requested production night mask.

6 Prices, residuals and information trees

Canonical CSV and explicitly mapped XML import preserve negative prices, reject duplicates/nonfinite labels and report gaps. Cache identity contains provider, dataset, zone, product, dates, vintage/revision and schema. Retrieval time does not establish historical publication time. Verified ENTSO-E and Terna API adapters are implemented and tested with injected offline transports. Authenticated downloads are user-run; this report reads completed local cache artifacts and does not make authenticated requests or infer account permissions. ENTSO-E supports native DA prices and load/generation/solar/wind forecasts [2, 1]. Terna supports load observations/forecasts, MGP load forecasts, MSD wind forecasts and national renewable actuals, preserving MW versus GWh [11, 14, 12, 15, 13]. IDA prices are not silently substituted. Terna fee-route ambiguity and the historical settlement adapter remain unresolved.

The GME API adapter implements authenticated PublicMarketResults requests for MGP and MI-A1/2/3 zonal prices, bounded JSON ZIP decoding and explicit native products. Documented-format offline fixtures exercise authentication, parser, cache and failure contracts. GME account access and user-authenticated historical downloads are still pending in the supplied evidence. No actual GME IDA price coverage or historical publication vintages are claimed.

The downloader plans daily Italy-local windows, reads caches before credentials, retries bounded transient failures, and reuses identical source requests. The default `month_first` order completes all configured datasets for one local month before advancing; `dataset_first` remains available. Scheduling is outside cache identity, so completed price and forecast days survive a restart. A pro-

cess already running retains its original plan until it is restarted. The offline `api-cache-status` command reports per-month and per-dataset coverage while acquisition continues. A ready month requires every configured daily payload; no-data receipts remain gaps. Readiness establishes payload presence, not historical information availability or experimental suitability. The cache catalog excludes unfinished writes and archived revisions, and model ingestion verifies raw and normalized checksums. Hidden terminal credentials are resolved once per process; token refresh uses the provider expiry. Current API configurations store requests in `data/api/market.duckdb`. The transactional store retains exact raw and normalized Parquet bytes, SQL records, request identities, retrieval clocks and snapshot provenance. Provider, dataset, zone and local day remain reusable metadata keys; existing folder caches can be migrated without another provider download. Experiment tables and plots remain grouped by run. Cache reads verify raw and normalized checksums; explicit refresh retains earlier snapshots. No-data receipts remain gaps and have a configurable lifetime. Historical publication and unverified Terna interval boundaries remain missing. Request schemas, credentials, subscriptions and actual coverage are recorded from the user's resulting caches when available.

Ridge uses training-only scaling, calendar cycles, local-clock one-day/seven-day lags and eligible prior auction prices. IDAs predict spreads from the preceding auction; lag joins preserve DST fold rather than shifting 96 rows. This causal interface omits regional fundamentals and local-to-regional proxies; the separate retrospective prototype below evaluates regional forecasts. Expanding chronological fits produce out-of-sample errors. Residual interfaces support shrunk Gaussian covariance, a labeled independent Gaussian, and whole-vector bootstrap; differently sized DST trajectories require separate calibration. The executed fixture has 288 synthetic records over 18 days and four selected products, not complete historical days. Four synthetic Ridge models each fitted 44 rows and retained 28 OOS residual rows; gaps remain explicitly reported.

6.1 Retrieved-revision DA holdout experiment

The configured experiment reads completed user-downloaded DA caches and preserves hourly and quarter-hour products as separate native-resolution studies. Training and evaluation follow delivery-day order, with training-only feature scaling. Prices retain the retrieved source revision; their historical publication and revision vintages are unverified. These results therefore measure retrospective price prediction, not executable historical trading or a causally calibrated scenario distribution.

Source coverage: 2025-01-01 through 2026-08-31; 608 complete days and 0 incomplete days. Native durations are never filled with synthetic quarter-hour observations.

15-minute study: 28596 training rows from 298 days; holdout 2026-08-02 through 2026-08-31, 30 observed days.

60-minute study: 5661 training rows from 236 days; holdout 2025-09-01 through 2025-09-30, 30 observed days.

Native min	Predictor	Holdout rows	MAE EUR/MWh	RMSE EUR/MWh
15	ridge	2880	17.897	25.592
15	previous day	2880	21.496	32.856
15	previous week	2880	22.836	31.011
60	ridge	720	9.536	12.896
60	previous day	720	11.485	18.666
60	previous week	720	9.685	13.523

Previous-day and previous-week prices provide seasonal baselines. These point-error summaries do not fit multiauction residual vectors, validate Gaussian noise or supply missing forecast publication vintages.

6.2 Regional forecast feature comparison

The configured experiment reads completed user-downloaded DA caches and preserves hourly and quarter-hour products as separate native-resolution studies. Training and evaluation follow delivery-day order, with training-only feature scaling. Prices retain the retrieved source revision; their historical publication and revision vintages are unverified. These results therefore measure retrospective price prediction, not executable historical trading or a causally calibrated scenario distribution.

Source coverage: 2025-01-01 through 2025-01-31; 31 complete days and 0 incomplete days. Native durations are never filled with synthetic quarter-hour observations.

This short regional-feature study compares price/calendar lags, added load, and added load/solar/onshore-wind forecasts. All models use identical complete training and holdout products. Regional MW forecasts are duration-averaged within native price products; incomplete coverage remains missing. Forecast revision and publication availability remain unverified. This is a pipeline prototype, not evidence of broad forecasting skill or executable trading.

60-minute study: 408 training rows from 17 days; holdout 2025-01-25 through 2025-01-31, 7 observed days.

Lowest MAE in this holdout: previous day. The holdout is too short to choose a generally superior forecasting method.

Native min	Predictor	Holdout rows	MAE EUR/MWh	RMSE EUR/MWh
60	ridge lags	168	13.024	15.315
60	ridge load	168	10.175	12.741
60	ridge load solar wind	168	9.643	11.965
60	previous day	168	8.855	11.458
60	previous week	168	18.905	30.361

Previous-day and previous-week prices provide seasonal baselines. These point-error summaries do not fit multiauction residual vectors, validate Gaussian noise or supply missing forecast publication vintages.

6.3 Rolling linear models and Gaussian trajectories

Each study selects its feature set, fitting method and penalty settings, level or previous-day difference target, and training window using an earlier validation block. Previous-day, previous-week and equal day/week forecasts also compete in selection. The selected specification is then fixed; coefficients refit weekly using only earlier delivery days and training-only scaling. A later calibration block estimates bias and error covariance; a delivery maturity gap separates it from the final chronological holdout. All candidates score identical complete-feature validation products. Hourly and quarter-hour regimes remain separate. Structure candidates include pooled regression and separate coefficients by local delivery hour, with quarter-hour observations retained inside their hour groups.

Candidate features include local-clock price lags, complete prior-day price summaries and profile blocks, rolling histories, calendar cycles and weekday/hour indicators. Regional studies add load, solar and wind forecasts or a separate residual-load specification; no realized target-day demand or generation is substituted for a forecast.

Additional fitting configurations compare ordinary least squares, recency-weighted ridge, fixed-scale Huber IRLS, lasso and elastic-net. Their family and hyperparameters compete on earlier validation MAE only; final-test results do not choose the estimator. OLS uses rank-safe least squares; recency weights enter both training scaling and the fitting loss. Iterative nonconvergence is reported and excludes that candidate, with no silent ridge substitution. The saved artifacts retain the actual fitting family.

Penalty magnitudes use distinct conventions: ridge minimizes $\sum_i w_i e_i^2 + \alpha\|\beta\|_2^2$ (unit weights for ordinary ridge), where recency weights sum to the training row count. Lasso and elastic-net use $\sum_i e_i^2/(2n) + \alpha r\|\beta\|_1 + \alpha(1-r)\|\beta\|_2^2/2$, with $r = 1$ for lasso. Identical numeric alpha values are therefore not equivalent penalties between these families. Huber uses a training-residual MAD scale and an unpenalized intercept; all scalers use the training slice only.

Prices: 2025-01-01 through 2026-08-31; 38711 native products. Earlier exploratory studies inspected September 2025 hourly and August 2026 quarter-hour outcomes. The final holdout is reserved from this study's selection; it is not an unseen prospective sample.

prices, 15 minutes: validation 2026-05-11–2026-06-07; calibration 2026-06-08–2026-07-19; final holdout 2026-07-21–2026-08-31.

prices, 60 minutes: validation 2025-06-10–2025-07-07; calibration 2025-07-08–2025-08-18; final holdout 2025-08-20–2025-09-30.

Gaussian scoring for prices, 15 minutes covers 42 days and 4032 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

The nominal 95 percent intervals cover only 89.7 percent of scored products in this holdout. This undercoverage is evidence that the current Gaussian uncertainty is not yet reliable in this regime.

Gaussian scoring for prices, 60 minutes covers 42 days and 1008 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

Fundamentals: 2025-01-01 through 2025-03-31; 2159 native products. Earlier exploratory studies inspected September 2025 hourly and August 2026 quarter-hour outcomes. The final holdout is reserved from this study's selection; it is not an unseen prospective sample.

fundamentals, 60 minutes: validation 2025-03-07–2025-03-13; calibration 2025-03-14–2025-03-23; final holdout 2025-03-25–2025-03-31.

Gaussian scoring for fundamentals, 60 minutes covers 5 days and 120 products; 2 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

Hourly_structure: 2025-01-01 through 2026-08-31; 38711 native products. Earlier exploratory studies inspected September 2025 hourly and August 2026 quarter-hour outcomes. The final holdout is reserved from this study's selection; it is not an unseen prospective sample.

This mean-model run ingested DuckDB snapshots from `data/api/market.duckdb`; 608 of 608 saved source snapshots have verified checksums. The report reads the run's saved provenance and frozen predictions.

hourly_structure, 15 minutes: validation 2026-05-11–2026-06-07; calibration 2026-06-08–2026-07-19; final holdout 2026-07-21–2026-08-31.

hourly_structure, 60 minutes: validation 2025-06-10–2025-07-07; calibration 2025-07-08–2025-08-18; final holdout 2025-08-20–2025-09-30.

Gaussian scoring for hourly_structure, 15 minutes covers 42 days and 4032 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

The nominal 95 percent intervals cover only 91.3 percent of scored products in this holdout. This undercoverage is evidence that the current Gaussian uncertainty is not yet reliable in this regime.

Gaussian scoring for hourly_structure, 60 minutes covers 42 days and 1008 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

Fitting_methods: 2025-01-01 through 2026-08-31; 38711 native products. Development comparison using the previously inspected July–August 2026 quarter-hour and August–September 2025 hourly periods. Estimator settings are selected by earlier validation MAE only. This comparison does not inspect or select from the September–October 2026 extension.

This mean-model run reused frozen features previously derived from DuckDB snapshots in `data/api/market.duckdb`; 608 of 608 saved source snapshots have verified checksums. The report reads the run's saved provenance and frozen predictions.

Frozen-feature source: `/home/rghetti/bess_pv_trading/outputs/linear_price_studies/hourly_structure/20261016T132545.918238Z/manifest.json`. The source manifest and feature SHA-256s are retained; this development comparison reuses its saved inputs without reopening the current database or adding newly downloaded extension labels.

fitting_methods, 15 minutes: validation 2026-05-11–2026-06-07; calibration 2026-06-08–2026-07-19; final holdout 2026-07-21–2026-08-31.

fitting_methods, 60 minutes: validation 2025-06-10–2025-07-07; calibration 2025-07-08–2025-08-18; final holdout 2025-08-20–2025-09-30.

Gaussian scoring for fitting_methods, 15 minutes covers 42 days and 4032 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

The nominal 95 percent intervals cover only 89.0 percent of scored products in this holdout. This undercoverage is evidence that the current Gaussian uncertainty is not yet reliable in this regime.

Gaussian scoring for fitting_methods, 60 minutes covers 42 days and 1008 products; 0 days are excluded for incomplete or unsupported native/DST grids. Its MAE/CRPS subset can therefore differ from the point-model comparison.

Study	Min	Validation-selected predictor	Val. MAE	Test MAE	Best test baseline
prices	15	day week average	17.106	17.498	day week average: 17.498
prices	60	ridge pooled, lags, alpha 100.0, 180d, delta day	10.273	8.788	day week average: 8.623
fundamentals	60	ridge pooled, residual_load, alpha 100.0, 28d, level	8.230	9.808	day week average: 11.768
hourly_structure	15	ridge by hour, profile, alpha 100.0, 90d, level	16.250	16.436	day week average: 17.498
hourly_structure	60	ridge pooled, lags, alpha 100.0, 90d, delta day	10.276	8.816	day week average: 8.623
fitting_methods	15	recency-weighted ridge by hour, lags, alpha 100.0, half-life 28.0d, 90d, level	16.164	17.431	day week average: 17.498
fitting_methods	60	day week average	10.666	8.623	day week average: 8.623

All MAEs are EUR/MWh. The best test baseline is descriptive and does not change the validation-selected forecast.

Study	Min	Cal. days	Error variance	Mean bias	95% cover	CRPS
prices	15	42	623.08	3.128	89.7	12.921
prices	60	42	165.83	-4.083	94.8	6.548
fundamentals	60	10	195.10	5.056	95.8	5.689
hourly_structure	15	42	500.59	4.742	91.3	12.259
hourly_structure	60	42	179.63	-0.120	96.6	6.331
fitting_methods	15	42	484.74	5.934	89.0	12.874
fitting_methods	60	42	172.30	-0.946	95.1	6.386

Error variance is mean product variance in $(\text{EUR/MWh})^2$; bias and CRPS are EUR/MWh, coverage is percent.

Sampled prices use $P_s = \widehat{\mu} + b + \varepsilon_s$, with $\varepsilon_s \sim \mathcal{N}(0, \Sigma)$. Bias b is estimated separately; the innovation mean is zero, while a finite Monte Carlo sample need not average to zero. Covariance comes from chronological out-of-sample forecast errors, not unconditional price variance, and shrinkage preserves temporal dependence. The Gaussian scenario adapter supplies native quarter-hour DA scenarios to the MILP information tree; hourly observations are not expanded into quarter-hour labels or dispatch prices. Current study artifacts remain explicitly ineligible for operational historical deployment. Missing IDA labels, verified publication vintages, PV eligibility and settlement semantics continue to prevent a validated historical trading backtest.

The actual-price benchmark builder in `forecasting/price_oracle.py` returns the same scenario-bundle and information-tree contracts with realized clearing-price curves. Each supplied PV member receives the identical actual price curve, with no Gaussian draws, while member probabilities are retained. PV member identities remain latent until completed-interval PV observations distinguish their histories; configurable telemetry delays can postpone that revelation. The oracle relaxes price information and is explicitly non-executable. Missing IDA auction prices are rejected, and native hourly observations are not interpolated into quarter-hour labels. For multiple PV members, a fair comparison uses the same physical constraints, member weights, observations and end-of-interval curtailment information for both forecasting policy and oracle. Unit and licensed analytical solve tests verify this construction; they do not establish historical trading profit.

6.4 Frozen-policy online Gaussian comparison

The following independent uncertainty runs reuse frozen out-of-sample point predictions. Inner chronological calibration days select the rolling error window and covariance shrinkage using the proper score recorded with each policy. Those policy settings remain fixed through the later test. Each online forecast may update error bias and covariance only from complete earlier delivery days that have matured before its issuance cutoff; a D-1 noon forecast therefore excludes all errors from delivery D-1. The static comparison keeps its original calibration bias and covariance fixed. Final-test outcomes never select policy hyperparameters. Earlier mature test errors do enter the explicitly online updates.

The linear point forecast is $\widehat{\mu}$; the Gaussian mean is $\widehat{\mu} + b$ and innovations remain $\varepsilon \sim \mathcal{N}(0, \Sigma)$. Point MAE below measures the unchanged linear model. Mean MAE measures the separate bias-corrected Gaussian mean, so the two need not agree. MAE and CRPS use EUR/MWh; interval and consecutive-product ramp coverage are percentages with a nominal target of 95.

Source mean study: hourly structure. Earlier exploratory studies inspected September 2025 hourly and August 2026 quarter-hour outcomes. The final holdout is reserved from this study's selection; it is not an unseen prospective sample.

The saved source-manifest copy matches the recorded SHA-256. Frozen calibration and test prediction files accompany the uncertainty run.

Source ingestion provenance identifies DuckDB `data/api/market.duckdb` and 608 checksum-verified snapshots. Gaussian fitting reads the point study's frozen prediction files.

15 minutes: selected 42-day window and shrinkage 0.3 using joint Gaussian negative log likelihood per product; policy-selection information ends 2026-07-19T22:00:00+00:00. Online scoring covers 42 days and 4032 native products. 100 seeded paths are saved for 2026-07-21.

For 15 minutes, rolling bias correction increases MAE from 16.436 to 17.195. Its uncertainty scores do not establish an improvement in the linear point forecast's accuracy.

The 15-minute online intervals cover 92.3 percent against a nominal 95 percent. This remaining undercoverage does not establish reliable tail probabilities or operational readiness.

60 minutes: selected 42-day window and shrinkage 0.1 using joint Gaussian negative log likelihood per product; policy-selection information ends 2025-08-18T22:00:00+00:00. Online scoring covers 42

days and 1008 native products. 100 seeded paths are saved for 2025-08-20.

Min	Error policy	Point MAE	Mean MAE	95% cover	CRPS	Ramp 95%
15	Static	16.436	17.140	91.27	12.259	98.72
15	Online	16.436	17.195	92.26	12.221	98.32
60	Static	8.816	8.644	96.63	6.331	99.17
60	Online	8.816	8.689	96.03	6.311	96.07

Source mean study: fitting methods. Development comparison using the previously inspected July–August 2026 quarter-hour and August–September 2025 hourly periods. Estimator settings are selected by earlier validation MAE only. This comparison does not inspect or select from the September–October 2026 extension.

The saved source-manifest copy matches the recorded SHA-256. Frozen calibration and test prediction files accompany the uncertainty run.

Source ingestion provenance identifies DuckDB `data/api/market.duckdb` and 608 checksum-verified snapshots. Gaussian fitting reads the point study's frozen prediction files.

15 minutes: selected 42-day window and shrinkage 0.3 using joint Gaussian negative log likelihood per product; policy-selection information ends 2026-07-19T22:00:00+00:00. Online scoring covers 42 days and 4032 native products. 100 seeded paths are saved for 2026-07-21.

For 15 minutes, rolling bias correction increases MAE from 17.431 to 18.351. Its uncertainty scores do not establish an improvement in the linear point forecast's accuracy.

The 15-minute online intervals cover 91.3 percent against a nominal 95 percent. This remaining undercoverage does not establish reliable tail probabilities or operational readiness.

60 minutes: selected 42-day window and shrinkage 0.1 using joint Gaussian negative log likelihood per product; policy-selection information ends 2025-08-18T22:00:00+00:00. Online scoring covers 42 days and 1008 native products. 100 seeded paths are saved for 2025-08-20.

For 60 minutes, rolling bias correction increases MAE from 8.623 to 8.624. Its uncertainty scores do not establish an improvement in the linear point forecast's accuracy.

The 60-minute online intervals cover 94.6 percent against a nominal 95 percent. This remaining undercoverage does not establish reliable tail probabilities or operational readiness.

Min	Error policy	Point MAE	Mean MAE	95% cover	CRPS	Ramp 95%
15	Static	17.431	18.000	88.99	12.874	98.42
15	Online	17.431	18.351	91.32	13.039	98.22
60	Static	8.623	8.648	95.14	6.386	99.07
60	Online	8.623	8.624	94.64	6.328	94.82

These are retrospective latest-retrieved revisions with unverified historical publication clocks. Operational eligibility remains false. The comparison does not provide missing IDA observations, causal forecast vintages, verified settlement inputs or historical trading-profit evidence.

6.5 Previously uninspected historical extension

A saved protocol fixes the selected mean specification, weekly coefficient refit schedule and Gaussian policy before examining the additional historical period. These dates are a previously uninspected extension of the research dataset, not future market outcomes observed prospectively. The check reads checksum-verified DuckDB snapshots, preserves the earlier feature history, and replays the original refit anchor before scoring new dates. It performs no model-family, feature, penalty or covariance-policy selection using the extension outcomes. Earlier eligible labels may refit coefficients; only complete earlier mature errors may update online uncertainty.

Scored extension: 2026-09-01 through 2026-10-16 (exclusive end 2026-10-17), following previously inspected 2026-08-31; 46 complete days and 4416 native quarter-hour products. Database: `data/api/market.duckdb`; 46 new source snapshots.

Frozen point specification: ridge by hour, profile, alpha 100.0, 90d, level. Gaussian window 42 days, shrinkage 0.3. Saved protocol creation (UTC): 2026-10-16T13:36:20.044724+00:00.

Saved-history immutability flag: True. The overlap replay compares 8160 prior products; maximum forecast difference 0.000000000 EUR/MWh, with tolerance 0.000000001. Identical cutoff and training-day audit: True. Copied protocol SHA-256 verified: True; ancillary source hashes pinned in the protocol: True.

Point predictor	MAE	RMSE	Selected MAE skill (%)
Frozen selected model	15.556	21.566	—
Previous day	21.841	32.635	28.78
Previous week	21.701	29.141	28.32
Equal day/week average	18.293	25.331	14.97

Point MAE/RMSE are EUR/MWh. Skill is one minus selected-model MAE divided by the named baseline's MAE; positive values favor the selected model.

Gaussian policy	Mean MAE	95% cover	CRPS	Ramp 95%	95% width
Static	16.158	92.44	11.739	98.47	81.219
Online frozen policy	15.378	94.25	11.249	98.72	82.056

Mean MAE includes the separately estimated Gaussian bias correction; it is distinct from linear point MAE. The static distribution retains its original calibration parameters; the online distribution updates earlier mature errors under the already-fixed policy. Interval width is EUR/MWh: closer nominal coverage must be assessed together with width and CRPS, rather than interpreted alone as better forecasts.

Online 95 percent intervals cover 94.2 percent in this extension, below nominal coverage. The new period does not establish reliable tail calibration.

Scored extension: 2026-09-01 through 2026-10-16 (exclusive end 2026-10-17), following previously inspected 2026-08-31; 46 complete days and 4416 native quarter-hour products. Database: `data/api/market.duckdb`; 46 new source snapshots.

Frozen point specification: recency-weighted ridge by hour, lags, alpha 100.0, half-life 28.0d, 90d, level. Gaussian window 42 days, shrinkage 0.3. Saved protocol creation (UTC): 2026-10-16T13:56:22.424172+00:00.

Saved-history immutability flag: True. The overlap replay compares 8160 prior products; maximum forecast difference 0.000000000 EUR/MWh, with tolerance 0.000000001. Identical cutoff and training-day audit: True. Copied protocol SHA-256 verified: True; ancillary source hashes pinned in the protocol: True.

Point predictor	MAE	RMSE	Selected MAE skill (%)
Frozen selected model	16.156	22.039	—
Previous day	21.841	32.635	26.03
Previous week	21.701	29.141	25.55
Equal day/week average	18.293	25.331	11.68

Point MAE/RMSE are EUR/MWh. Skill is one minus selected-model MAE divided by the named baseline's MAE; positive values favor the selected model.

Gaussian policy	Mean MAE	95% cover	CRPS	Ramp 95%	95% width
Static	16.379	90.87	11.895	98.33	79.620
Online frozen policy	16.063	95.24	11.667	98.88	87.408

Mean MAE includes the separately estimated Gaussian bias correction; it is distinct from linear point MAE. The static distribution retains its original calibration parameters; the online distribution updates earlier mature errors under the already-fixed policy. Interval width is EUR/MWh: closer nominal coverage must be assessed together with width and CRPS, rather than interpreted alone as better forecasts.

Historical publication and revision vintages remain unverified, and operational eligibility remains false. This extension tests statistical prediction under a frozen protocol; it does not establish a causal historical trading backtest or actual strategy profit.

The finite tree branches at results publication, keeps later descendants unresolved, and groups identical sibling observed vectors to avoid leaking latent labels. DA innovations belong to its level equation; later innovations belong to conditional spread equations. No second cross-auction residual is added. Known results remain fixed. Current weather is frozen until a new publication; no probabilistic forecast-revision model is supplied. The synthetic replay publishes one perfect deterministic AC forecast; it is not a weather-skill experiment. The earlier prepared-input adapter also connects complete exported AC forecasts and supplied point-price forecasts to the same builder. It checks forecast source, fit and availability cutoffs, one farm/member, review exclusions, equipment and an explicit settlement study assumption. Its contract was tested on fixtures; no real price feed was supplied to this adapter. This route supports initial pre-DA planning, not historical replay. Imbalance prices use an explicit synthetic point estimate. Solver-independent scenario contracts accept joint PV arrays. The new artifact boundary below accepts explicit weighted PV members; actual meteorological ensemble acquisition and calibration remain unexecuted.

6.6 Artifact-to-stochastic optimization input boundary

The executable `optimization/forecast_inputs.py` interface provides `build_forecast_inputs` for in-memory canonical tables and `load_forecast_inputs` for saved forecast/model artifacts. The `optimize` command routes an explicit `optimization.forecast_inputs` configuration to the same `OptimizationInput` contract and MILP builder. This boundary accepts explicitly weighted PV members, supplied price means or fitted DA/conditional IDA spread artifacts, with Gaussian calibration when uncertainty is requested. It forms an independent price-times-PV product measure using common price draws across PV members; no calibrated dependence between weather/PV and market prices is claimed. Member identities remain latent until completed PV observations distinguish histories.

Every submitted quantity executes in full under the price-taking assumption and becomes a fixed commitment immediately. A committed quantity awaiting price publication remains fixed in every residual scenario. Recorded known clearing curves remain fixed; publication books quantity times actual clearing price once in the immutable priced ledger. The legacy accepted-trade field names denote already executed, priced quantities. Prior committed cash is accounted for externally; the optimization objective measures incremental new trading and dispatch profit. Residual reoptimization uses an exact remaining product grid and an explicit boundary physical state. IDA3 data and calibration cover afternoon products only; coefficients outside its false trading mask are inert placeholders, not observed clearing prices. Source, fit and availability cutoffs are checked, and retrospective artifacts require an explicit study opt-in. Matching fixture execution is recorded in the supplied passing JUnit suite. The schema and assumptions are documented in `forecast_optimization_inputs.md`. These contracts do not supply absent GME labels, verified historical publication vintages, settlement economics or operational farm forecasts. No actual-data economic optimization backtest is established

by the fixtures.

7 Formulation matching the code

For leaf s with mass p_s, delivery interval t with duration h_t hours, control owner $n(t,s)$ and auction owner $m(a,s)$, charge c, discharge d, curtailment u and signed grid power x use MW. Internal energy e and financial trades v use MWh. The battery's charge mode z is binary and shared by node:

$$x_{ts} = g_{ts} - u_{ts} + d_{t,n(t,s)} - c_{t,n(t,s)}, \tag{1}$$

$$e_{t+1,s} = r_t e_{ts} + \eta_c c_{t,n(t,s)} h_t - d_{t,n(t,s)} h_t/\eta_d, \tag{2}$$

$$0 \le c_{tn} \le \bar{c} z_{tn}, \qquad 0 \le d_{tn} \le \bar{d}(1 - z_{tn}), \tag{3}$$

$$\underline{e} \le e_{ts} \le \bar{e}, \quad -\bar{x}^{in} \le x_{ts} \le \bar{x}^{out}, \quad 0 \le u_{ts} \le g_{ts}. \tag{4}$$

Efficiencies are converter times cell efficiencies, or a validated combined convention; losses are never applied twice. Retention is $(1-\ell)^{h_t}$. Initial state timestamps equal the residual grid start; terminal energy is a fixed equality in every leaf. Power/energy reachability is prechecked; remaining infeasibility is reported with optional LP/IIS diagnostics.

Financial positions and imbalances obey

$$q_{ts} = q_t^{priced} + q_t^{unpriced} + \sum_a v_{a,m(a,s),t}, \qquad b_{ts} = x_{ts} h_t - q_{ts}. \tag{5}$$

Only eligible open auctions receive variables. Both per-trade and cumulative commercial positions are bounded. Submitted quantities execute in full and remain fixed while awaiting price publication. There is no acceptance uncertainty. Their unknown cash flow is a scenario constant excluded from residual optimization and identified in diagnostics. On publication, quantity times the actual clearing price is booked once in the priced ledger. The internal pending and accepted field names distinguish unpriced and priced commitments, respectively. Opposite financial trades incur no battery efficiency loss.

The exact signed settlement uses $b = b^+ - b^-$ and a bounded binary direction split, so simultaneous positive and negative components cannot create fictitious profit:

$$R_s^{imb} = \sum_t (\pi_{ts}^+ b_{ts}^+ - \pi_{ts}^- b_{ts}^-), \tag{6}$$

$$C_s^{imb} = \rho \sum_t (b_{ts}^+ + b_{ts}^-), \tag{7}$$

$$C_s^{age} = k \sum_t (c_{t,n(t,s)} + d_{t,n(t,s)}) h_t, \tag{8}$$

$$\max \quad \sum_s p_s \left[\sum_{a,t} \pi_{ts}^a v_{a,m(a,s),t} + R_s^{imb} - C_s^{imb} - C_s^{age} - C_s^{other} \right]. \tag{9}$$

The fixed absolute-imbalance penalty is the requested study deterrent; it does not establish historical law or guarantee compliance. Synthetic imports have no extra grid fee; excluded taxes/brokerage remain explicit. Throughput is AC charge plus discharge, with cumulative accounting carried across events. The optional `xu_segmented` model implements Xu's Eq. (5), (7), (8)–(15) and appendix Eq.

(37)–(40) [16]. It uses a convex depth stress function and equal internal-energy segments:

$$\Phi(\delta) = \alpha\delta^{\beta}, \quad \bar{e}_j = E/J, \tag{10}$$

$$k_j = \frac{RJ}{\eta_d E}\left[\Phi(j/J) - \Phi((j-1)/J)\right], \tag{11}$$

$$C_s^{age,Xu} = \sum_{t,j} h_t k_j d^j_{t,n(t,s)}, \tag{12}$$

$$e^j_{t+1,s} = e^j_{ts} + h_t\left(\eta_c c^j_{t,n(t,s)} - d^j_{t,n(t,s)}/\eta_d\right). \tag{13}$$

Segment powers sum to the physical battery command and segment energies sum to internal energy. Explicit priority binaries select the appendix's shallow-first allocation for both charging and discharging; these are cycle-memory segments, not fixed absolute-SoC buckets. Charge-only half cycles have zero aging charge; discharge half cycles incur the full same-depth cost. No half factor is applied. Initial occupancy is assigned once, and the segment vector is carried through committed commands, execution and residual optimization. Independent numerical replay and rainflow comparisons check the segment policy and cost.

The demonstrated Xu configuration uses eight segments, a separately assumed EUR300,000 pack replacement cost, and the paper's illustrative $\alpha = 0.000524, \beta = 2.03$. These are not calibrated farm-battery parameters. Xu mode currently requires zero self-discharge; the throughput baseline remains available. The priority representation adds $2(J-1)$ binaries per physical control node and interval. Details and exact run evidence are in `xu_degradation.md`.

One Gurobi deterministic equivalent holds shared decisions; its related-model multiple-scenario facility is not used [6]. Solve results distinguish optimality, time limits with/without incumbents, infeasibility and unboundedness [7]. Extraction requires an incumbent. A separate numerical validator recomputes balances, modes, positions, settlement, node sharing and weighted profit. The configured relative optimality tolerance is 10^{-4} (0.01%); no absolute MIP gap is configured, following the user's general rule.

8 Execution and observed results

Events order forecast publications, auction results and completed observations before coincident submissions/physical controls, then settlement. A computation buffer shifts submission cutoffs. Only the current auction quantity is submitted; future modeled adjustments are discarded until their actionable event. Submission fixes the fully executed quantity immediately. Price publication books its actual cash once, refreshes conditional future price forecasts and triggers residual reoptimization while preserving committed quantities and physical commands. These forecast objective values remain separate from realized ledger cash flow. The synthetic results handler rebuilds conditional descendant prices from all published actual curves at each publication. Its diagnostic residual solve submits no additional quantity and replaces no command already fixed for the current physical interval. The solve trace records the publication trigger, known-price checks, booked quantity and actual cash increment. The four-auction fixture checks these updates, single cash booking and carried state across an intrainterval publication. Saved replay totals below retain the run-specific configuration and timestamps of their listed manifests. Completed interval averages never enter the command for their own interval. State and cumulative aging carry forward. At a cutoff within an already committed interval, its known battery command projects the next boundary without reading its final PV average. The safety response is idealized at interval-average resolution and corrections are recorded; the synthetic support has perfect PV and required none. This is not high-frequency hardware validation or continuous overlapping multi-day simulation.

The final JUnit evidence records 616 passed, 0 failed, 0 errors and 0 skipped tests. Historical market backtests, causally verified historical scenario calibration and rear-side bifacial validation were

not executed. Authenticated acquisition remains user-run; any supplied DA experiment evidence describes a retrieved-revision retrospective price study. Xu aging was executed on analytical cycles, nested-cycle/rainflow comparisons, shared-scenario controls and a full 96-product auction replay. The ordinary synthetic day compares four different policy definitions on the same realization and constraints. Equal integrated-policy profits on this case do not establish value from stochastic optimization or justify modifying the economics. The oracle uses future truth and is non-executable. Realized operating profit is distinct from initial expected profit and excludes any salvage credit.

Day	Products	Policy	Profit EUR	Replay s	Max error
2025-10-02	96	deterministic_sequential	483.9815	3.321	5.68e-14
2025-10-02	96	deterministic_integrated	669.0537	3.556	1.48e-12
2025-10-02	96	stochastic_integrated	669.0537	7.091	1.48e-12
2025-10-02	96	oracle	724.2398	3.413	1.25e-12
2025-10-02	96	deterministic_integrated (Xu)	542.6887	15.797	1.93e-12
2025-10-26	100	stochastic_integrated	675.0537	7.616	1.82e-12
2026-03-29	92	stochastic_integrated	663.0537	6.666	1.36e-12

The initial stochastic ordinary-day model had 15088 variables, 2736 binaries and 22400 constraints, with 16 leaves. Measured build/solve seconds were 0.5831 and 0.2318; status optimal, bound 570.4132409539403 EUR, gap 0.0. All values are synthetic, not financial performance estimates. Model size/runtime for the one-leaf configuration is saved separately as a reproducible scaling reference, if executed.

9 Reproducibility, artifacts and limitations

Activate the existing `bess` environment and run from the repository; no installation step is needed. The application is executable with `PYTHONPATH=src python -m bess_pv_trading`. Concrete commands:

```
PYTHONPATH=src python -m bess_pv_trading validate-config \
  --config configs/farms_nord.yaml
PYTHONPATH=src python -m bess_pv_trading inspect-data \
  --config configs/farms_nord.yaml
PYTHONPATH=src python -m bess_pv_trading prepare-analysis-data \
  --config configs/farms_nord.yaml
PYTHONPATH=src python -m bess_pv_trading forecast-pv \
  --config configs/farms_nord.yaml
PYTHONPATH=src python -m bess_pv_trading run-synthetic \
  --config configs/synthetic_small.yaml
PYTHONPATH=src python -m bess_pv_trading import-market-data \
  --config configs/synthetic_prices.yaml
PYTHONPATH=src python -m bess_pv_trading train-price-models \
  --config configs/synthetic_prices.yaml
PYTHONPATH=src python -m bess_pv_trading api-cache-status \
  --config configs/api_history.yaml
PYTHONPATH=src python -m bess_pv_trading experiment-api-prices \
  --config configs/price_experiment.yaml
PYTHONPATH=src python -m bess_pv_trading study-linear-prices \
  --config configs/price_linear_study.yaml
PYTHONPATH=src python -m bess_pv_trading study-linear-prices \
  --config configs/price_linear_methods.yaml
```

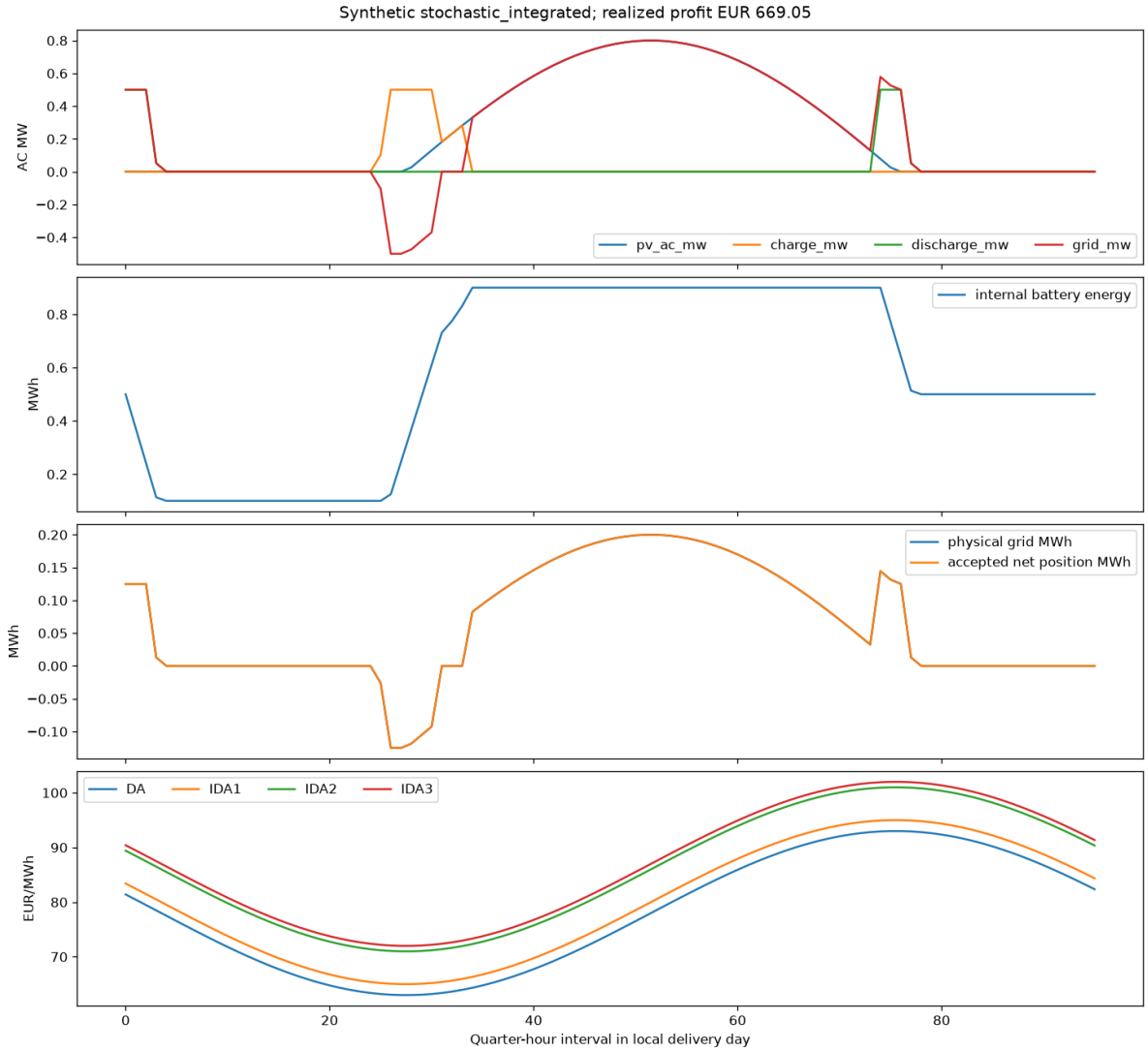


Figure 1: Executed synthetic stochastic policy, AC powers, energy, positions and prices.

```
PYTHONPATH=src python -m bess_pv_trading calibrate-price-scenarios \
  --config configs/price_gaussian_adaptive.yaml
PYTHONPATH=src python -m bess_pv_trading optimize \
  --config configs/stochastic_prepared.example.yaml
PYTHONPATH=src python -m pytest -q --junitxml=docs/generated/tests.xml
PYTHONPATH=src python -m bess_pv_trading report-implementation \
  --config configs/report.yaml
```

The long commands above may be line-wrapped by the PDF viewer; README contains copyable commands. Report generation reads the actual manifests and JUnit file; it does not rerun tests or synthesize missing evidence. Run directories cannot overwrite existing experiment results. The real-data example remains exploratory and writes explicit ineligibility rather than inventing missing physical inputs. The prepared stochastic optimization configuration is a schema template: supply its documented forecast artifacts and equipment/state assumptions before solving.

Selected numerical evidence paths, relative to the repository:

- `outputs/trading/20261016T101049649174Z/farm_1/2025-10-02/manifest.json`
- `outputs/trading/20261016T103126975966Z/farm_1/2025-10-02/manifest.json`
- `outputs/trading/20261016T101049576294Z/farm_1/2025-10-26/manifest.json`
- `outputs/trading/20261016T101049735769Z/farm_1/2026-03-29/manifest.json`
- `outputs/analysis/farm1_20251001/manifest.json`
- `outputs/pv/farm1_20251001/manifest.json`
- `outputs/market_price_fixture/models/training_manifest.json`
- `docs/generated/tests.xml`

The historical eligible intersection for executable strategy validation is presently empty because complete causal auction and verified settlement inputs remain unavailable. Retrieved DA prices support retrospective price analysis separately. PV eligibility and upstream review would further restrict it. BESS/POI hardware, actual coordinates, tracker geometry, rear response and PV-output conversion/calibration remain assumed. No independent historical strategy validation or meteorological ensemble calibration is claimed. Any Gaussian CRPS and coverage above concern retrieved-revision price studies and the explicitly reported scored subset. Four farms must continue to be evaluated separately. The source documentation and unresolved meanings are detailed in `dataset_contract.md`, `assumptions.md`, `formulation.md` and `sources.md`.

TeX validation status: compiled-successfully. PDF visual inspection status is separately recorded in the JSON evidence.

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